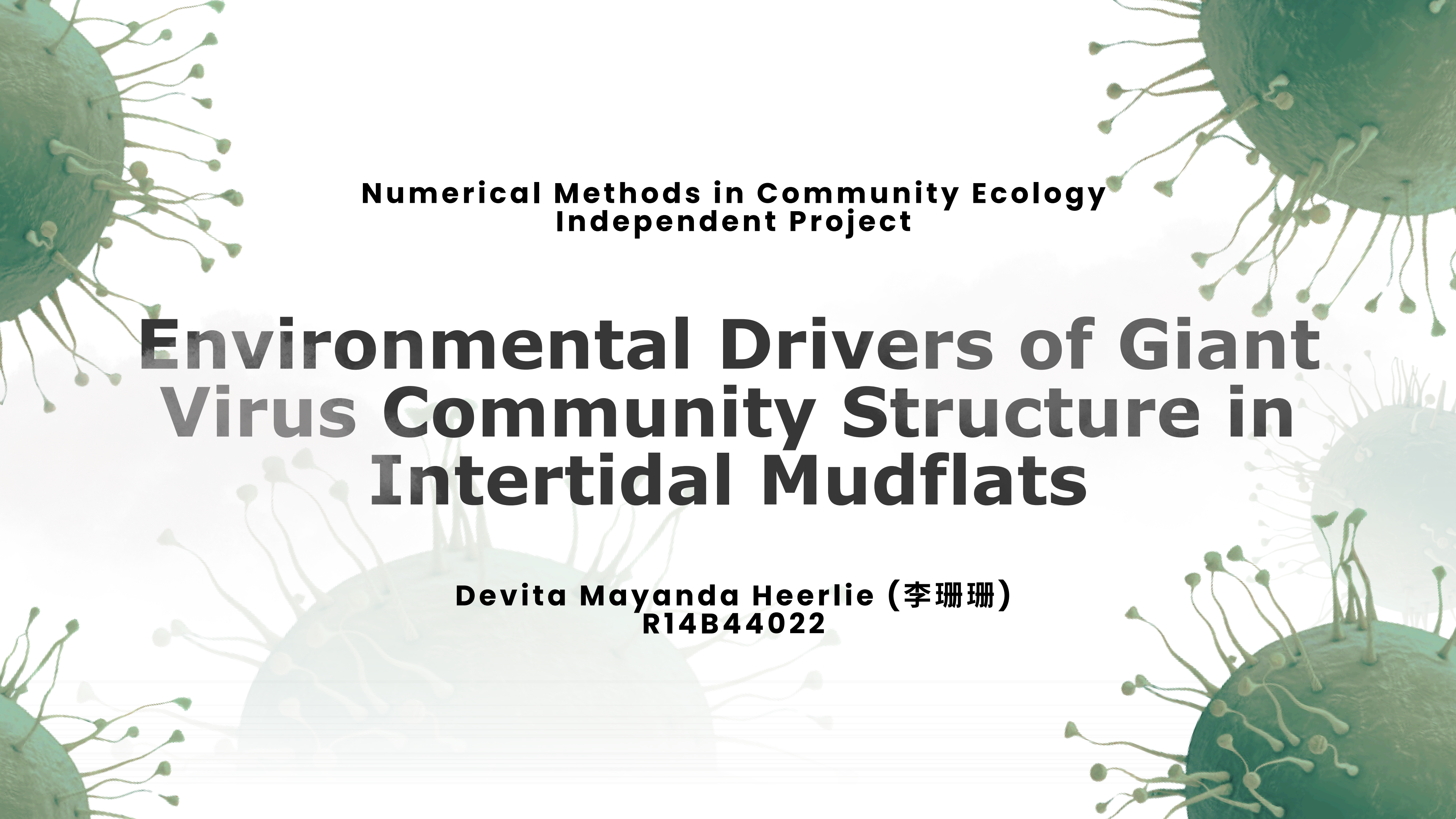




**Numerical Methods in Community Ecology
Independent Project**

Environmental Drivers of Giant Virus Community Structure in Intertidal Mudflats

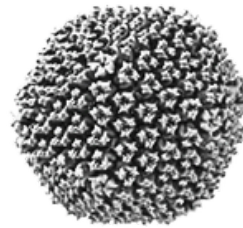
**Devita Mayanda Heerlie (李珊珊)
R14B44022**



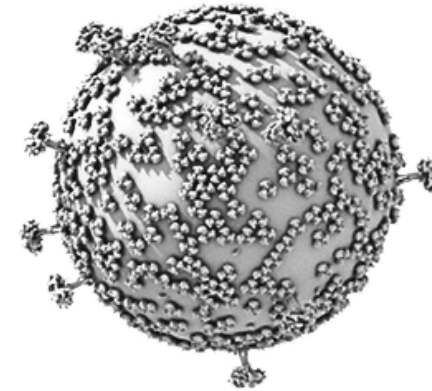
Polio Virus
~30 nm diameter



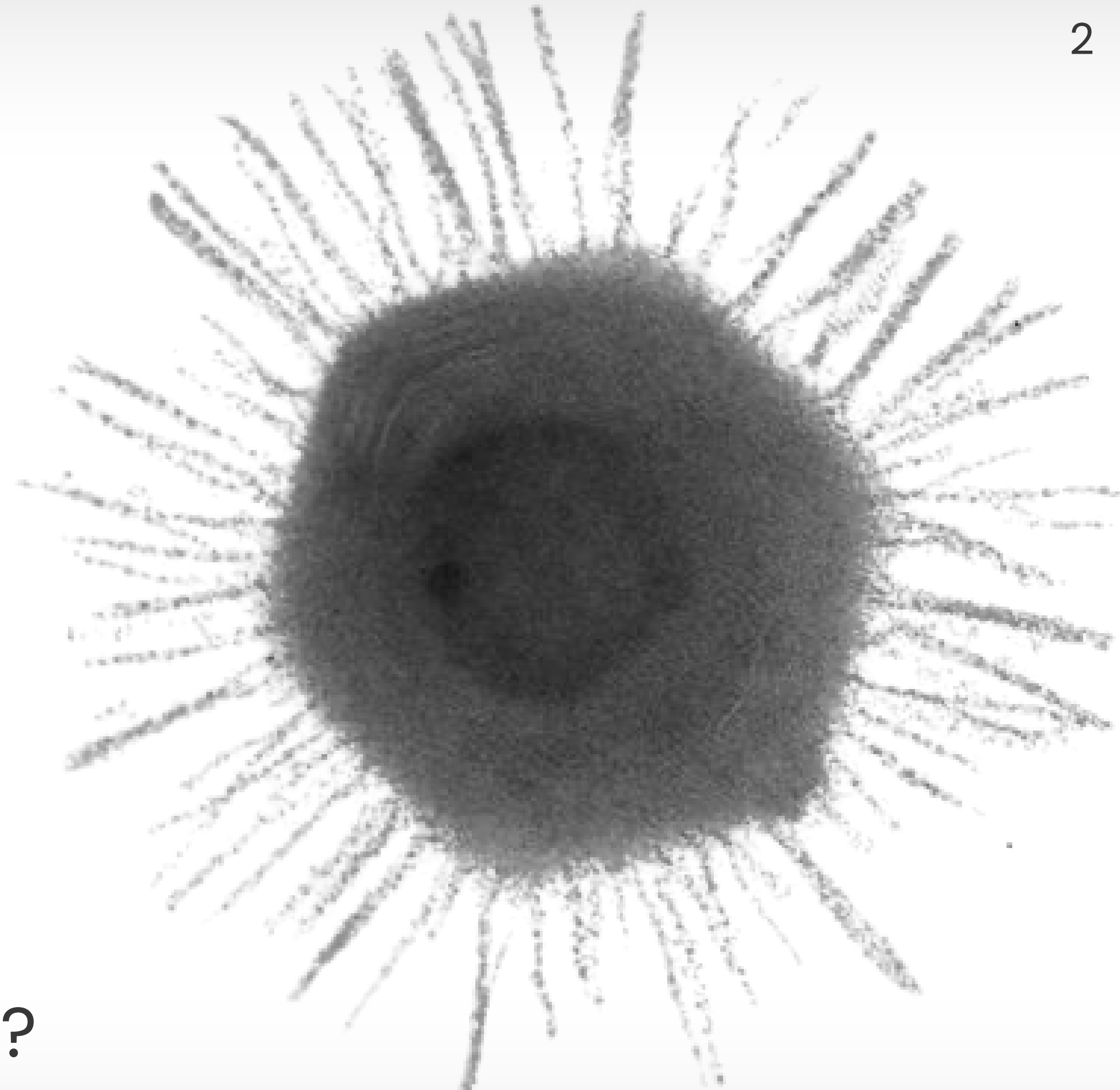
Zika Virus
~45 nm diameter



Adenovirus
~90 nm diameter



HIV 1
~120 nm diameter

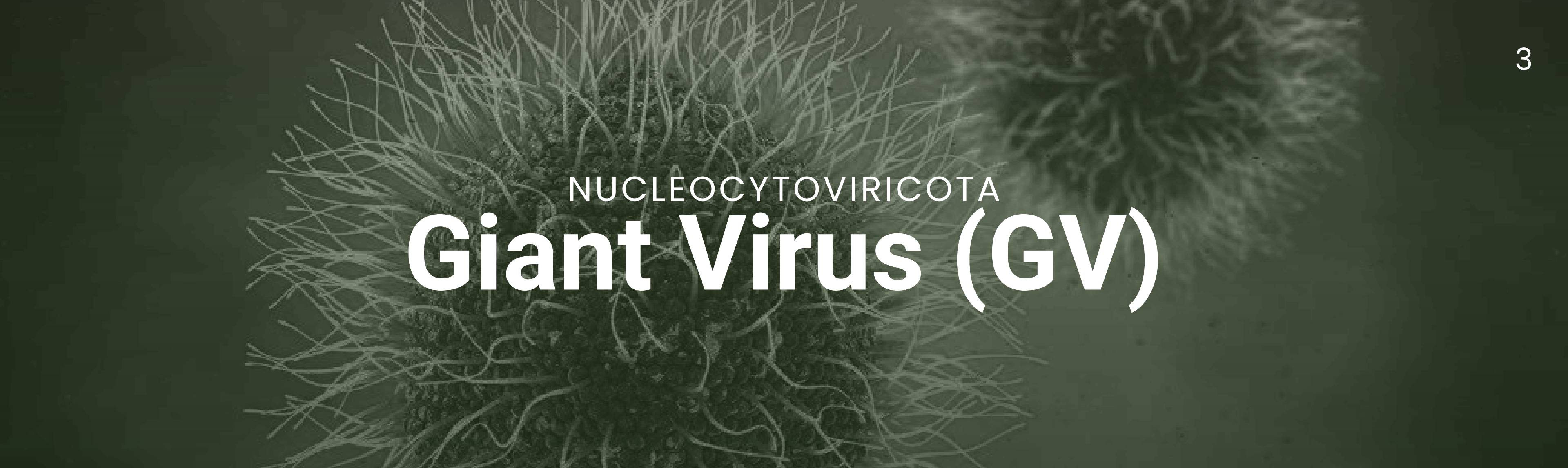


Acanthamoeba polyphaga mimivirus (APMV)
~750 nm diameter

Nucleocytoviricota

Giant Virus

The name refers to the size?



NUCLEOCYTOVIRICOTA

Giant Virus (GV)

01

LARGE SIZE AND GENOME

GV have large particle (100 - 2500 nm) size and large genome (100 - 2500 kb)

02

GIANT VIRUS DIVERSITY

GVs are highly diverse and abundant in various environments.

03

CURRENT GAP

Environmental parameters shaping giant virus community structures are poorly understood

Research Question?

Do specific viral taxonomic clusters correlate with specific environmental variables?

Research Objectives?

- Identify the key environmental factors driving giant virus communities through forward selection, and model the direct relationships between specific viral taxonomic clusters
- Identify the corresponding environmental constraints by coupling multi-ordination analysis with hierarchical classification

Data Sources

Nat Commun. 2026 Mar 24;17:4358. doi: [10.1038/s41467-026-71095-7](https://doi.org/10.1038/s41467-026-71095-7)

nature communications



Article

Unveiling the biodiversity of large DNA viruses in intertidal mudflats via metagenomics

Mengzhi Ji ¹, Yan Li¹, Mengqi Wang¹, Xia Liu¹, Xiaofan Gong ¹ & Qichao Tu ^{1,2,3}

¹Institute of Marine Science and Technology, Shandong University, Qingdao, China

²Qingdao Key Laboratory of Ocean Carbon Sequestration and Negative Emission Technology, Shandong University, Qingdao, China

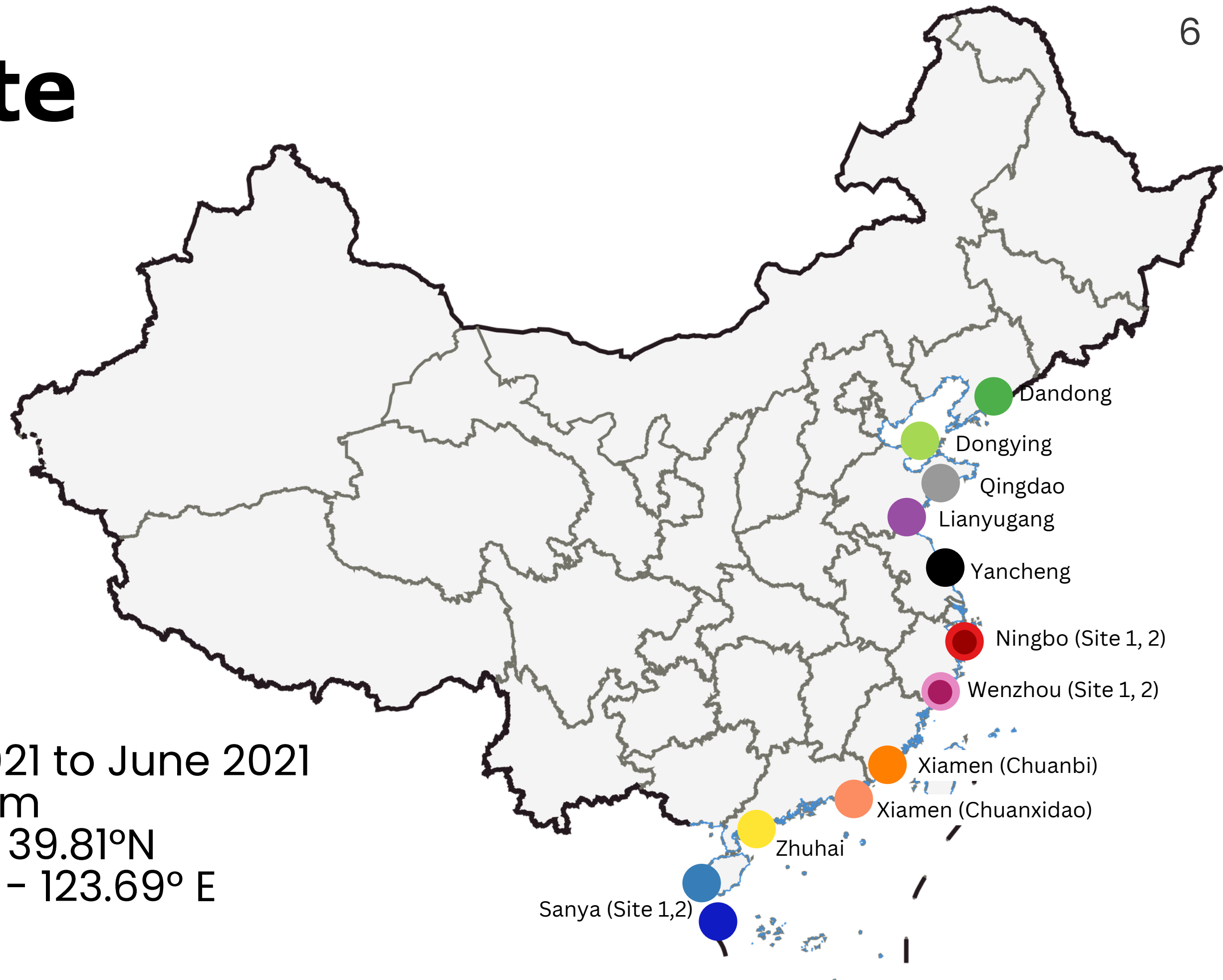
³Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai, China

Sampling Site



14 Sampling Site

Sampling Date April 2021 to June 2021
 Sampling Depth 0 – 15 cm
 Latitude 18.27° – 39.81°N
 Longitude 109.68° – 123.69° E



Experimental Design

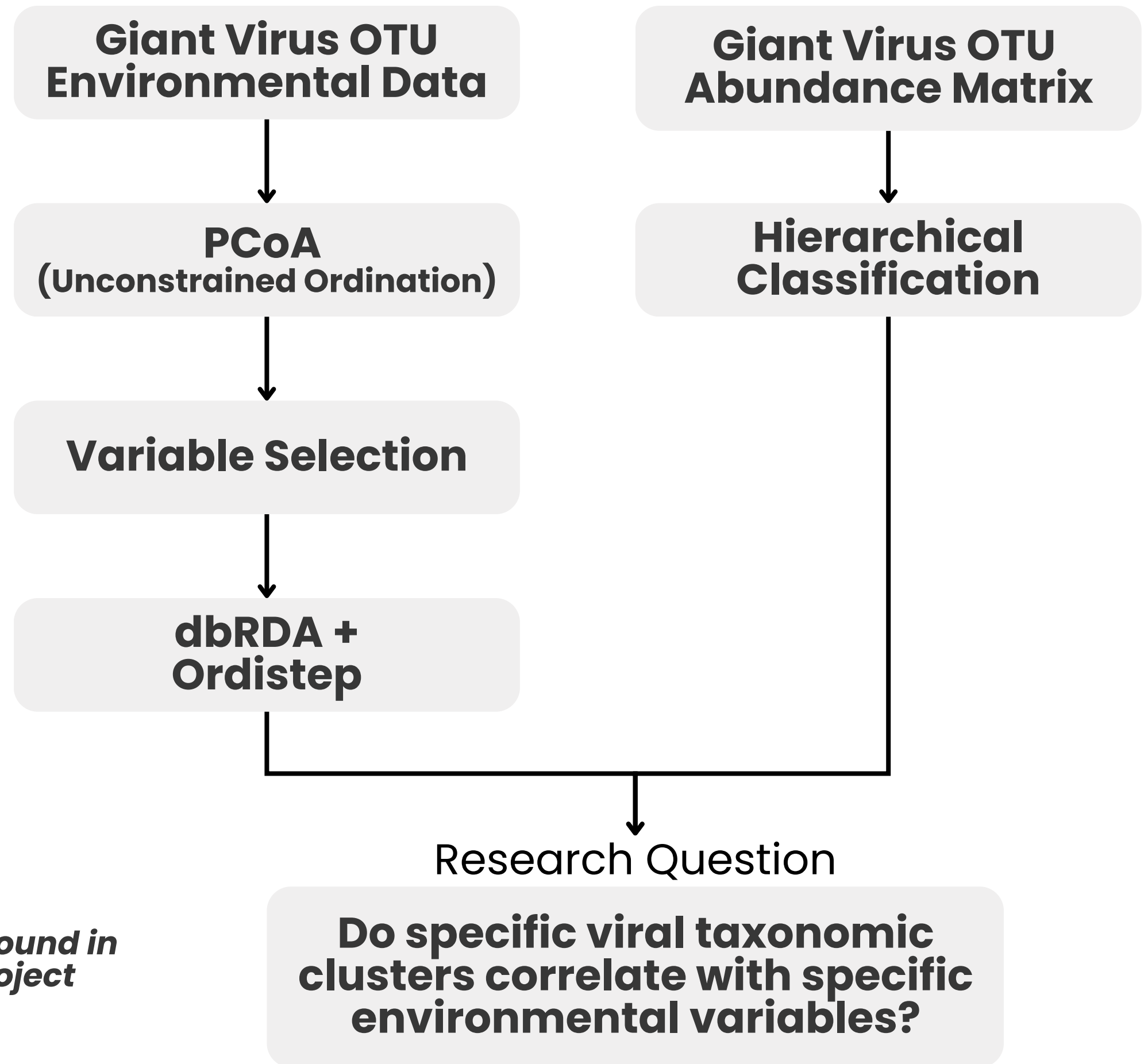
Environmental Data

- Physical factors (MAT, Moisture, pH)
- Salinity/Ions (salinity, SO₄)
- Nutrients (NH₄, NO₃, NO₂, TN, TP)
- Carbon pools (TOC)

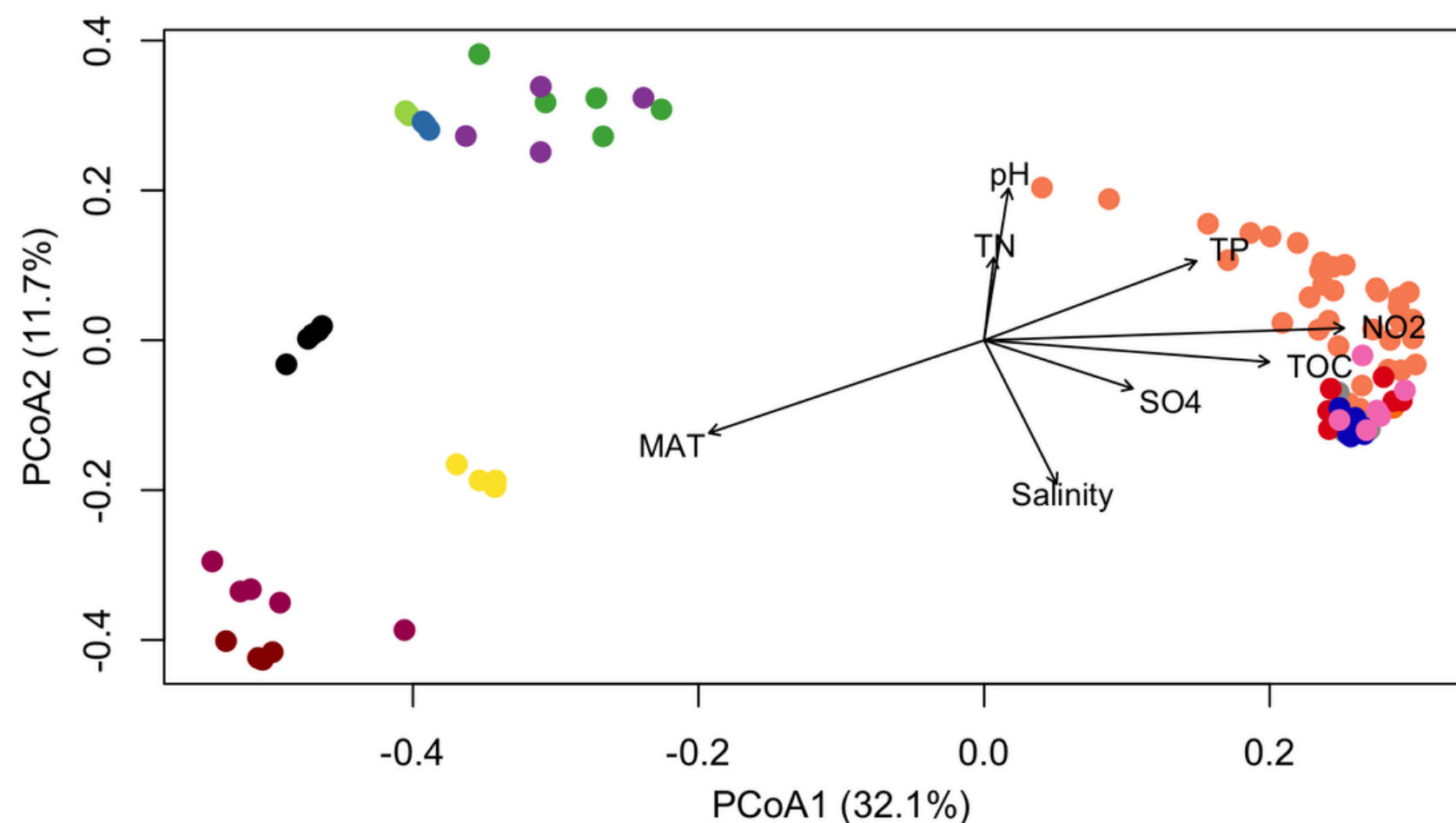
Giant Virus OTU

- Abundance matrix of GVMAGs OTUs

All of the data and scripts done in this project could be found in <https://github.com/mesosupp/NumEcolIndependentProject>



PCoA with Environmental Data

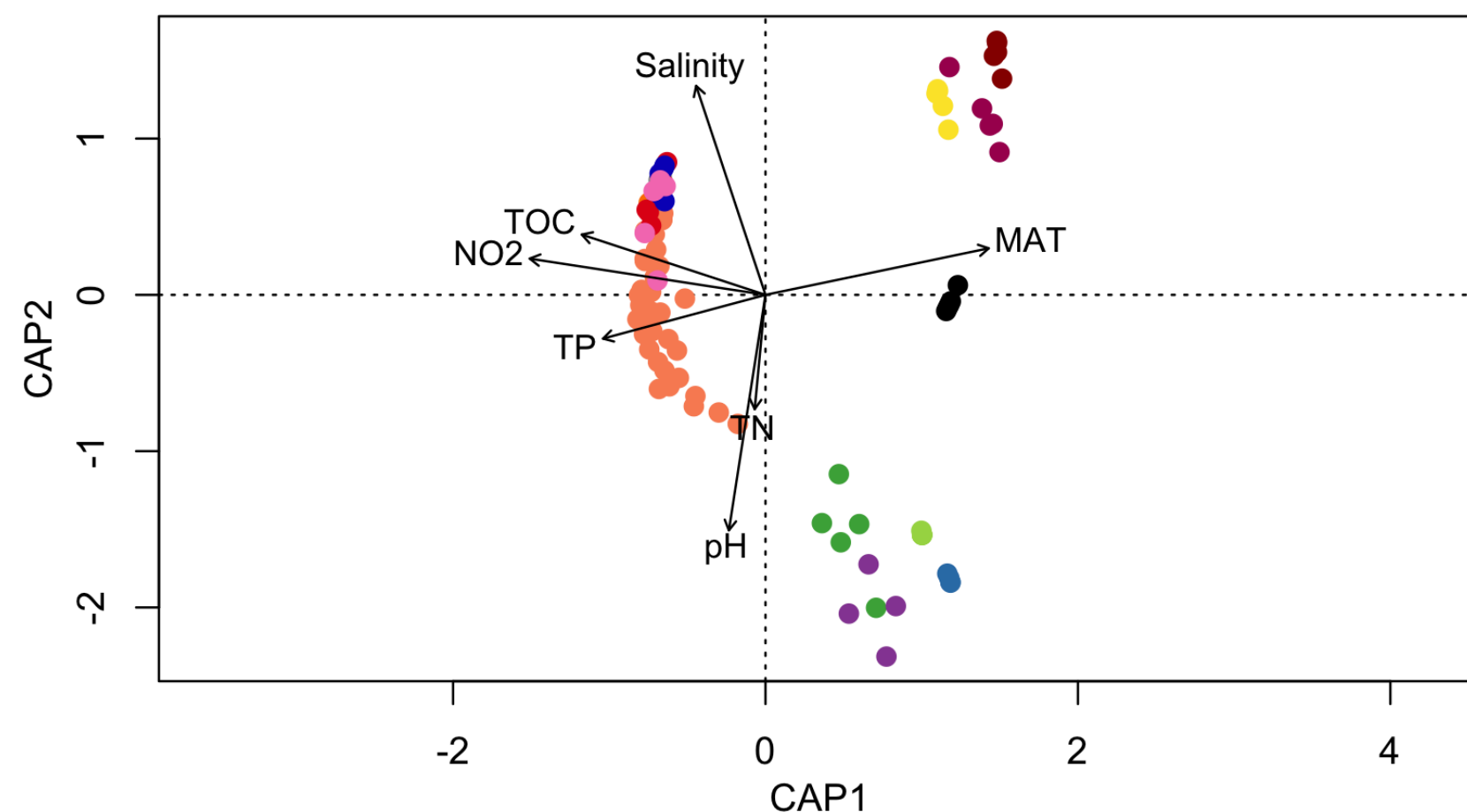


dbRDA with Selected Variables

KEY FINDINGS

- The forward selection procedure identified **seven key environmental drivers**
- Giant virus communities forming **a core cluster** consist of multiple sites, alongside **scattered clusters**
- The dbRDA explains the distribution patterns seen in the PCoA, confirming that **local environmental filters override geographic distance**

dbRDA with Forward Selection



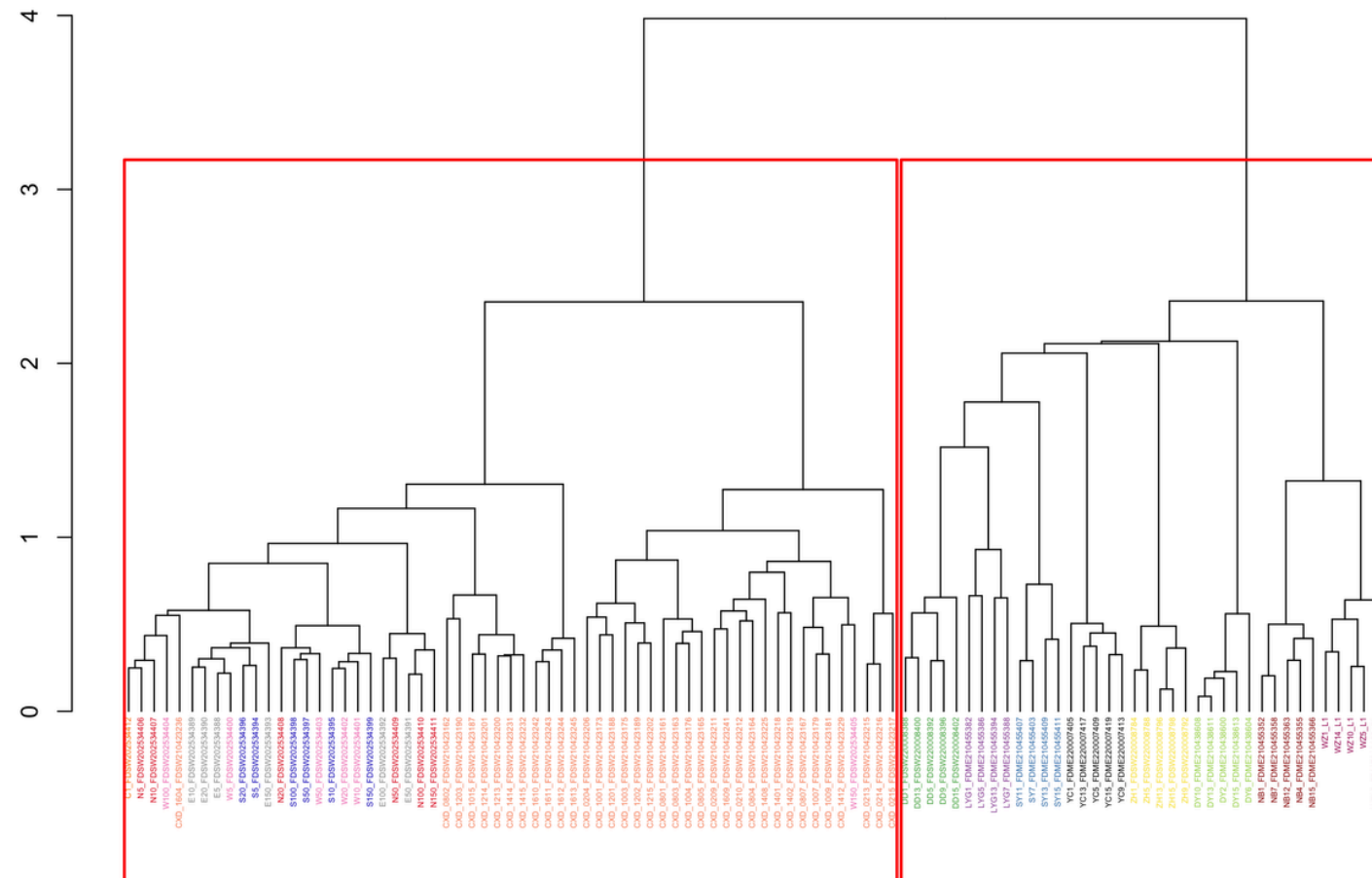
Sampling Sites

● C	Xiamen (Chuanbi)	● NB	Ningbo (Site 2)
● CXD	Xiamen (Chuanxidao)	● S	Sanya (Site 1)
● DD	Dandong	● SY	Sanya (Site 2)
● DY	Dongying	● W	Wenzhou (Site 1)
● E	Qingdao	● WZ	Wenzhou (Site 2)
● LYG	Lianyugang	● YC	Yancheng
● N	Ningbo (Site 1)	● ZH	Zhuhai

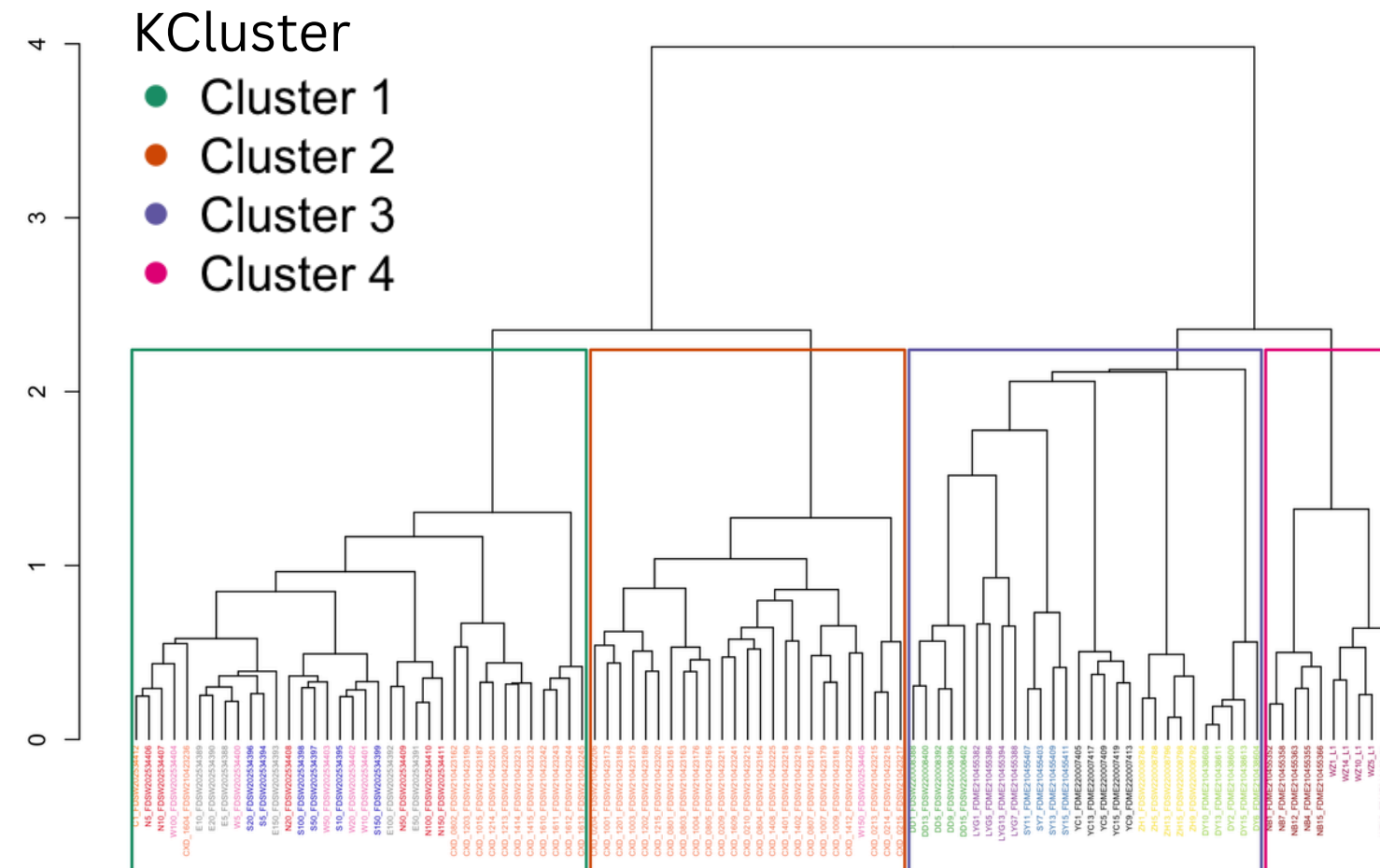
HIERARCHICAL CLASSIFICATION

Ward's Clustering Comparison

Ward's Clustering (k=2)



Ward's Clustering (k=4)



Sampling Site

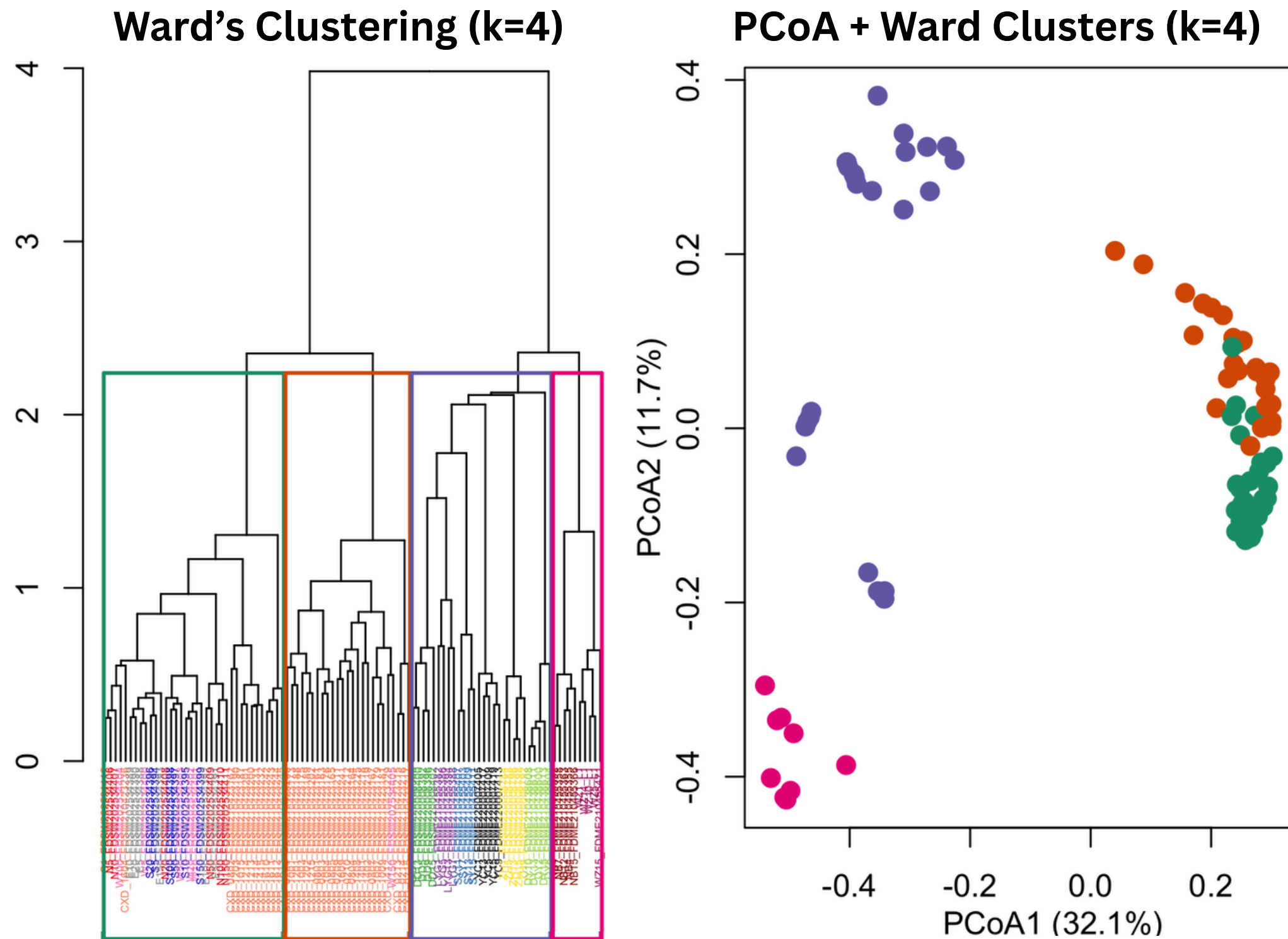
- C Xiamen (Chuanbi)
- CXD Xiamen (Chuanxidado)
- DD Dandong
- DY Dongying
- E Qingdao
- LYG Lianyungang
- N Ningbo (Site 1)
- NB Ningbo (Site 2)
- S Sanya (Site 1)
- SY Sanya (Site 2)
- W Wenzhou (Site 1)
- WZ Wenzhou (Site 2)
- YC Yancheng
- ZH Zhuhai

Ward's methods generated a well-resolved structure of the clusters

- k=2 separated the core clusters from regional outliers
- k=4 clusters matched the continuous ordination space perfectly
- k=5 (overclustering) fragments the core clusters.

HIERARCHICAL CLASSIFICATION

Spatial Distribution of Plots



The figure overlays the clusters by Ward's hierarchical classification directly onto the PCoA ordination space (k=4)

Sampling Site

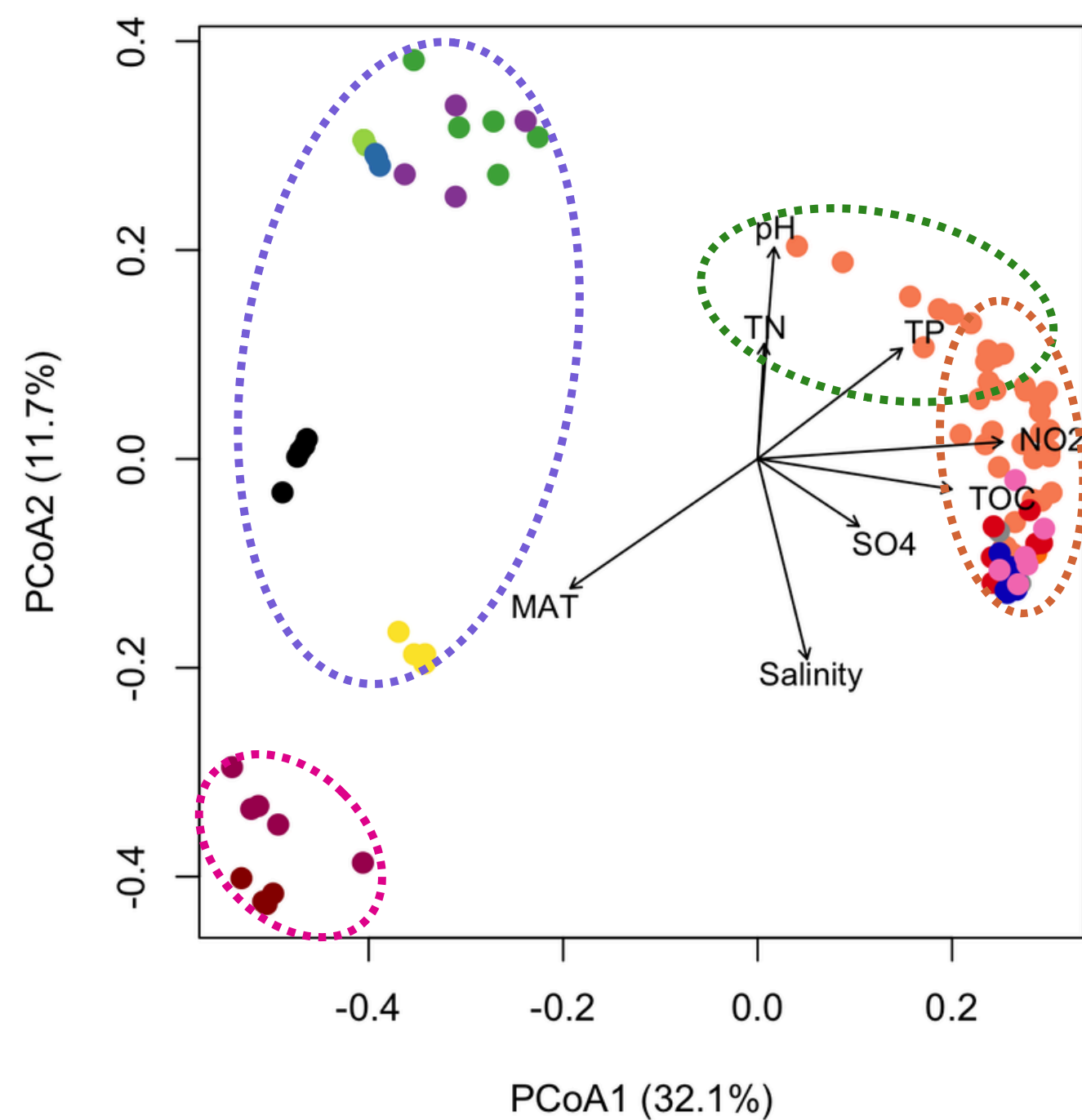
- **C** Xiamen (Chuanbi)
- **CXD** Xiamen (Chuanxidao)
- **DD** Dandong
- **DY** Dongying
- **E** Qingdao
- **LYG** Lianyugang
- **N** Ningbo (Site 1)
- **NB** Ningbo (Site 2)
- **S** Sanya (Site 1)
- **SY** Sanya (Site 2)
- **W** Wenzhou (Site 1)
- **WZ** Wenzhou (Site 2)
- **YC** Yancheng
- **ZH** Zhuhai

KCluster

- Cluster 1
- Cluster 2
- Cluster 3
- Cluster 4

Multi Covariate Analysis with Clustering

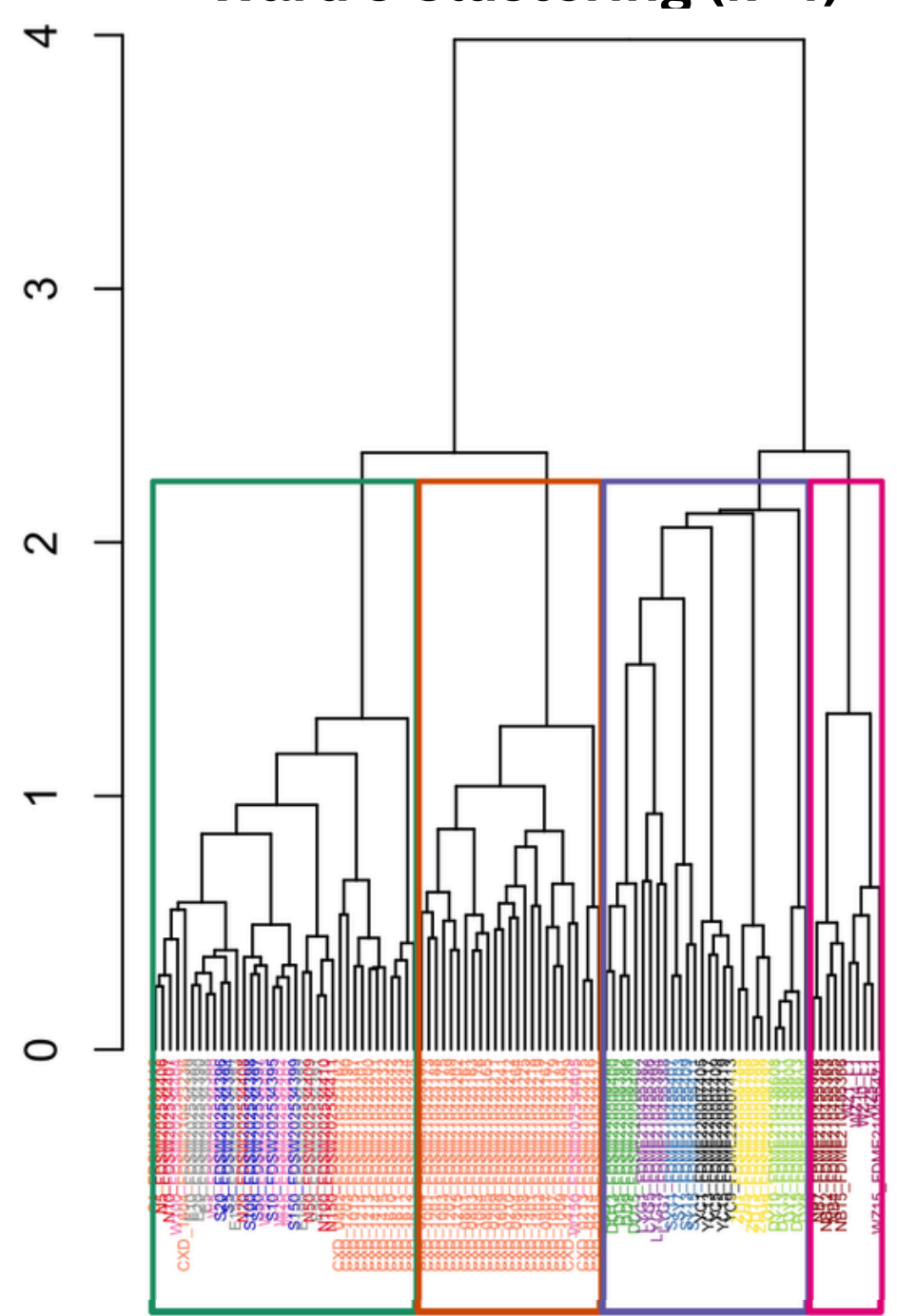
PCoA + Environmental Vectors



Sampling Site

- | | | | |
|-------|---------------------|------|------------------|
| ● C | Xiamen (Chuanbi) | ● NB | Ningbo (Site 2) |
| ● CXD | Xiamen (Chuanxidao) | ● S | Sanya (Site 1) |
| ● DD | Dandong | ● SY | Sanya (Site 2) |
| ● DY | Dongying | ● W | Wenzhou (Site 1) |
| ● E | Qingdao | ● WZ | Wenzhou (Site 2) |
| ● LYG | Lianyugang | ● YC | Yancheng |
| ● N | Ningbo (Site 1) | ● ZH | Zhuhai |

Ward's Clustering (k=4)



KCluster

- | |
|-------------|
| ● Cluster 1 |
| ● Cluster 2 |
| ● Cluster 3 |
| ● Cluster 4 |

Research Question

Do specific viral taxonomic clusters correlate with specific environmental variables?

Cluster 1 (Green)

- Variables: pH, TN, TP
- Sites: Xiamen

Cluster 2 (Orange)

- Variables: TOC, NO2, SO4, TP
- Sites: Sanya (1), Ningbo (2), Xiamen, Zhuhai, Qingdao
- Baseline environment for giant virus

Cluster 3 (Purple)

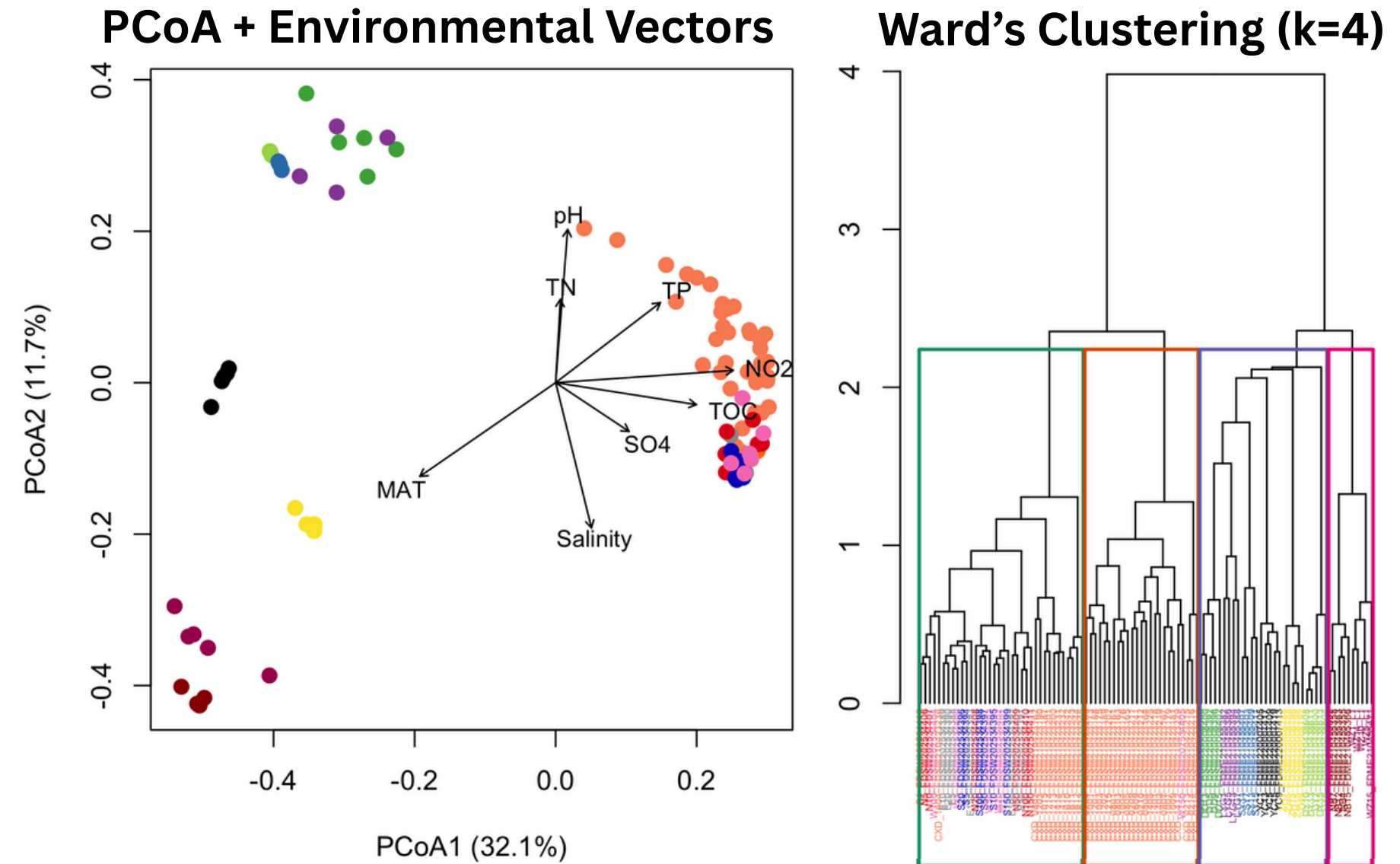
- Variable: MAT
- Sites: Dandong, Dongying, Sanya (2), Lianyugang, Yancheng, Zhuhai

Cluster 4 (Pink)

- Variable: Salinity
- Site: Ningbo (1)

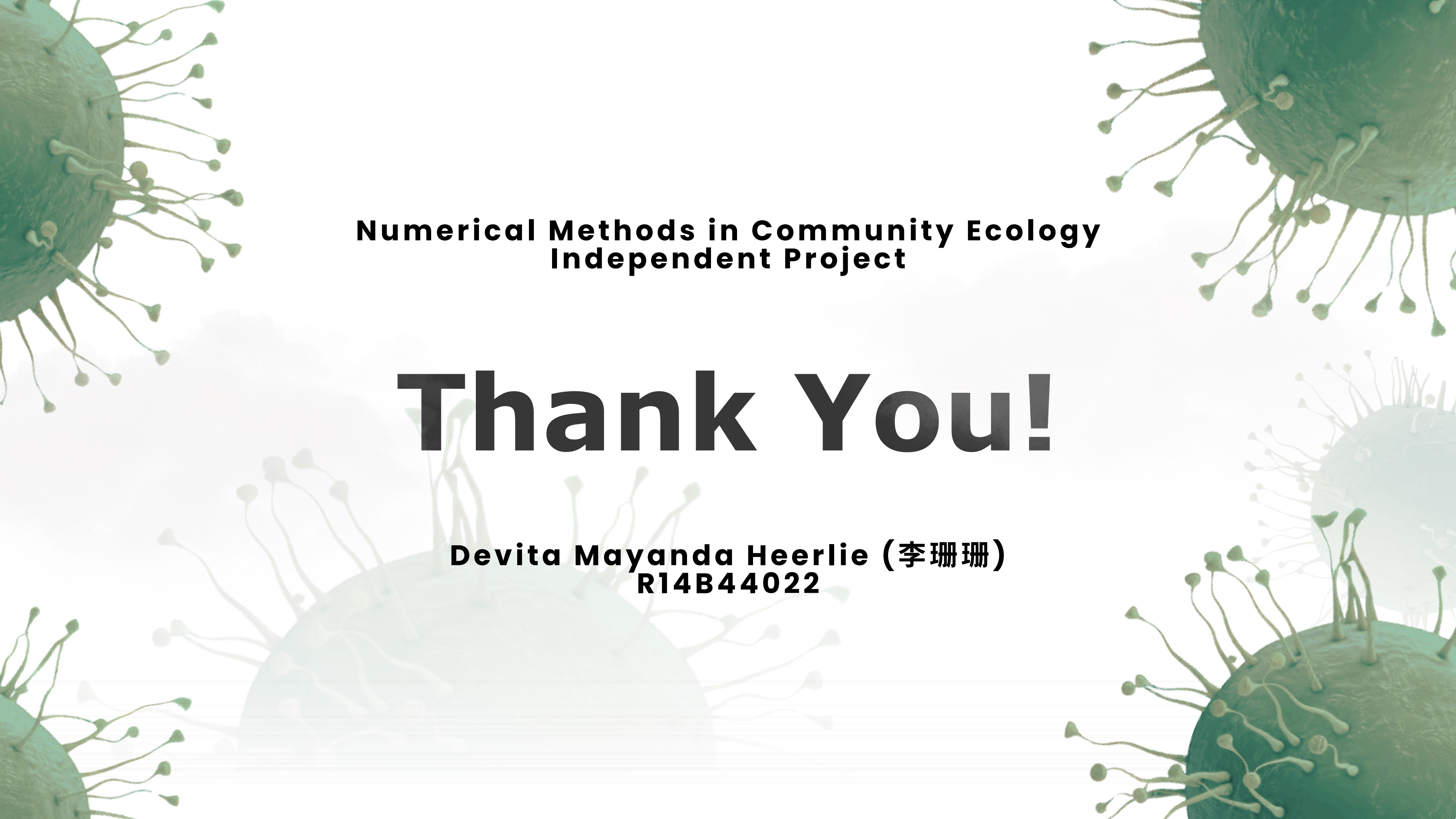
Conclusion

- Coastal giant virus biogeography is strictly structured by deterministic environmental sorting.
- Identified a baseline community with the same environmental factors across diverse geographic provinces.
- Extreme local environmental thresholds (salinity and MAT) forming the "satellite clusters," overriding geographic distance barriers.
- Establishes a definitive, statistically verified baseline for understanding the distribution and niche boundaries of Nucleocytoviricota (giant viruses) in dynamic coastal zones.



Limitations

- Unmeasured environmental variables
- Not lineage-specific, but more to community analysis



**Numerical Methods in Community Ecology
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Thank You!

**Devita Mayanda Heerlie (李珊珊)
R14B44022**